



TEAMS SYNC

A cross-college platform that helps students find teammates by skill, verify real project work via GitHub, and assemble hackathon teams fast.

Find Teammates

GitHub Verified

Match & Form Teams

Introduction

What TeamSync is

TeamSync is a web-based cross-college platform built for students who need a teammate their own institute doesn't have, for a hackathon or a project.

Every student gets a verified profile with tech stack, GitHub repos, and interests, and a ranked matching engine that surfaces complementary teammates.



Proposed for UCS503P at TIET — a 12-week engineering build piloted across at least two colleges around an upcoming hackathon.



Tech Stack Profiles

Structured tags for stack, domain & experience



GitHub Integration

Verifiable signal of real, shipped work



Cross-College Network

Find teammates your own college lacks

Project Background

The problem TeamSync addresses



Fragmented discovery

Teammate search happens through scattered contacts, WhatsApp/Discord groups, or generic posts — with no shared, searchable channel.



No cross-college visibility

A student needing a Flutter developer and one who fits that need at another college have no platform to find each other.



Low trust & verification

Open forums can't confirm a profile genuinely belongs to a verified college student.



Poor skill discoverability

No structured way to filter people by tech stack, availability, or complementary skills.

Core Workflow

From profile to team, in four steps



1. Register & Build Profile

Sign up with a verified college email and build a profile with tech stack, GitHub links, and interests.



2. Post or Browse Listings

Post a “looking for teammates” requirement, or browse and search existing listings by skill.



3. Ranked Matching & Connect

The matching engine surfaces ranked candidates; accepted connections unlock in-app chat.



4. Team Workspace

Once a team forms, members get a shared workspace for task allocation and status tracking.

Methodology

Engineering approach — workflow over novel algorithms



Verified Profile & GitHub Link

Sign-up is restricted to verified college email domains, with GitHub linkage as proof of real, shipped work.



Complementary-Skill Matching

Heuristic scoring combines skill complementarity, interest overlap, experience, and availability with configurable weights.



Three-Tier Web Architecture

React.js + Tailwind CSS frontend, Node.js/Express backend, and MongoDB Atlas for stateless, scalable APIs.



Agile Delivery with CI/CD

Weekly iterations with automated build, test, and lint, plus one-command deployment to staging.

Timeline

12-week delivery across weekly iterations

Wks 1–2	Requirement Analysis & Literature Review	Problem statement, SRS draft
Wks 3–4	System Design	ER diagram, architecture diagram, UI mockups
Wks 5–7	Backend Development	Working REST APIs, DB integration
Wks 6–8	Frontend Development	Responsive web app connected to backend
Wks 8–9	Matching Engine & Team Workspace	Complementary-skill matching, shared workspace
Wk 10	Testing	Unit, integration, and pilot UAT test reports; bug fixes
Wks 11–12	Deployment, Presentation & Submission	Hosted demo, final report, proposal defense

Risks & Mitigations

Key risks identified for the pilot



Cold Start

Too few users to match. Run the pilot around a scheduled hackathon so demand and supply arrive together.



Cross-College Onboarding

Start with two colleges rather than an open rollout, to keep verification and support manageable.



Evaluation Ambiguity

Instrument timestamps and define listing, match, and connection states before development begins.



Operational Issues

Use staging early and ensure CI/CD produces repeatable, safe deployments.

Results

As a proposal, success will be measured against these pilot targets — recruiting 15–25 students across at least two colleges around a hackathon.



Time-to-First-Match

Median match time under 24 hours, from listing to first relevant candidate.



Cross-College Match Rate

Share of successful matches formed between students at different colleges.



Verification & Conversion

Profile verification, GitHub linkage, and connection-to-team conversion rates.